

MODULE 4: INTERNAL CPU ORGANIZATION AND IMPLEMENTATION

4.1 Overview of CPU Organization

The Central Processing Unit (CPU) is the brain of the computer. It consists of three main internal components:

- Arithmetic Logic Unit (ALU) – Performs arithmetic (addition, subtraction, etc.) and logical (AND, OR, NOT, etc.) operations.
- Control Unit (CU) – Directs the flow of data and instructions; generates control signals.
- Register Set – Small, fast storage locations inside the CPU used to hold temporary data, addresses, and instructions.

These components communicate with each other via internal buses and with memory and I/O via the system bus (address, data, control).

4.2 Detailed Register Set of a Basic Computer

A basic computer (as described in many textbooks) contains several registers, each with a specific purpose. The number and types vary by architecture, but the following are common.

4.2.1 Accumulator Register (AC)

- The accumulator is a general-purpose register that holds one operand for ALU operations and also stores the result.
- In many simple computers (and some modern microcontrollers), arithmetic and logic instructions implicitly use the accumulator as both source and destination.
- Example: ADD X adds the value at memory location X to the accumulator, leaving the result in the accumulator.
- Size: Typically equal to the word length (e.g., 8, 16, or 32 bits).

4.2.2 Address Register (AR) – also called Memory Address Register (MAR)

- Holds the address of a memory location to be read from or written to.
- Connected directly to the address bus.
- Size: Equal to the width of the memory address (e.g., 16 bits for 64 KB memory, 32 bits for 4 GB memory).

4.2.3 Data Register (DR) – also called Memory Data Register (MDR) or Memory Buffer Register (MBR)

- Holds the data that has been read from memory or the data to be written to memory.
- Connected directly to the data bus.
- Size: Equal to the word length (e.g., 16 or 32 bits).

4.2.4 Instruction Register (IR)

- Holds the current instruction being executed.
- The control unit decodes the opcode field of the IR to determine what action to take.
- Size: Equal to the instruction length (may be larger than the word length in some architectures).

4.2.5 Program Counter (PC)

- Holds the address of the next instruction to be fetched.
- After each instruction fetch, the PC is incremented (by the instruction length in bytes or words) or updated by jump/branch instructions.
- Size: Same as the address register (AR).

4.2.6 Temporary Register (TR)

- A scratchpad register used by the control unit to hold intermediate values temporarily.
- For example, during a multiplication or division operation, the TR may store partial results.
- Size: Same as the word length.

4.2.7 Input Register (INPR)

- Holds data received from an input device (e.g., keyboard, mouse) before it is transferred to the accumulator or memory.
- Size: Typically 8 bits for character input (ASCII/Unicode code point).
- The CPU reads the INPR when an input device signals that data is ready.

4.2.8 Output Register (OUTR)

- Holds data that is to be sent to an output device (e.g., display, printer).
- The output device controller reads from the OUTR when it is ready to transmit.
- Size: Typically 8 bits for character output.

4.2.9 Other Registers (Depending on Architecture)

- Stack Pointer (SP) – Points to the top of the stack in memory (used for function calls, interrupts).
- Status/Flag Register (SR or FLAGS) – Holds condition flags (Zero, Carry, Overflow, Sign, etc.) – covered in Module 2.
- General-Purpose Registers (R0, R1, ..., Rn) – Used by the programmer for any purpose (common in RISC architectures).

4.3 Common Bus System

In a basic computer, connecting every register to every other register would require too many wires. Instead, registers are connected through a common internal bus.

4.3.1 How the Common Bus Works

- All registers are connected to a shared set of lines (the bus).
- Only one register can output its contents to the bus at any given time (via tri-state buffers).
- Multiple registers can input from the bus simultaneously (by enabling their load signal).
- Selection of which register outputs is controlled by the control unit using selection lines (like a multiplexer).

4.3.2 Typical Registers on the Common Bus

In a basic computer, the following registers are typically connected to the internal common bus:

- Accumulator (AC)
- Address Register (AR)
- Data Register (DR)
- Instruction Register (IR)
- Program Counter (PC)
- Temporary Register (TR)
- Input Register (INPR) – often connected via the bus to AC
- Output Register (OUTR) – often connected from AC via the bus

Memory (RAM) is also connected to the same bus, but memory access is controlled separately via the MAR and MDR.

4.3.3 Example: Transfer from PC to AR

To fetch an instruction, the control unit:

1. Enables the PC output onto the bus.
2. Enables the AR input to load the value from the bus.
3. The bus now carries the address from PC to AR.
4. Then the control unit performs a memory read using the address now in AR.

This bus-based approach reduces wiring complexity but requires careful timing (only one register drives the bus at a time).

4.4 The Instruction Cycle

The CPU repeatedly executes the instruction cycle for each instruction. It consists of four phases:

4.4.1 Fetch Phase

- Purpose: Retrieve the next instruction from memory into the IR.
- Steps:

1. Transfer PC to AR ($PC \rightarrow \text{bus} \rightarrow \text{AR}$).
 2. Perform memory read. Memory places the instruction into DR (MDR).
 3. Transfer DR to IR ($DR \rightarrow \text{bus} \rightarrow \text{IR}$).
 4. Increment PC ($PC \leftarrow PC + 1$, or $PC + \text{instruction length}$).
- After fetch, the IR contains the instruction to be executed.

4.4.2 Decode Phase

- The control unit decodes the opcode field of the IR.
- It determines the operation to perform and the addressing mode(s) used.
- It may also compute the effective address (e.g., adding an index register).
- The decode phase is often combined with fetch or execute.

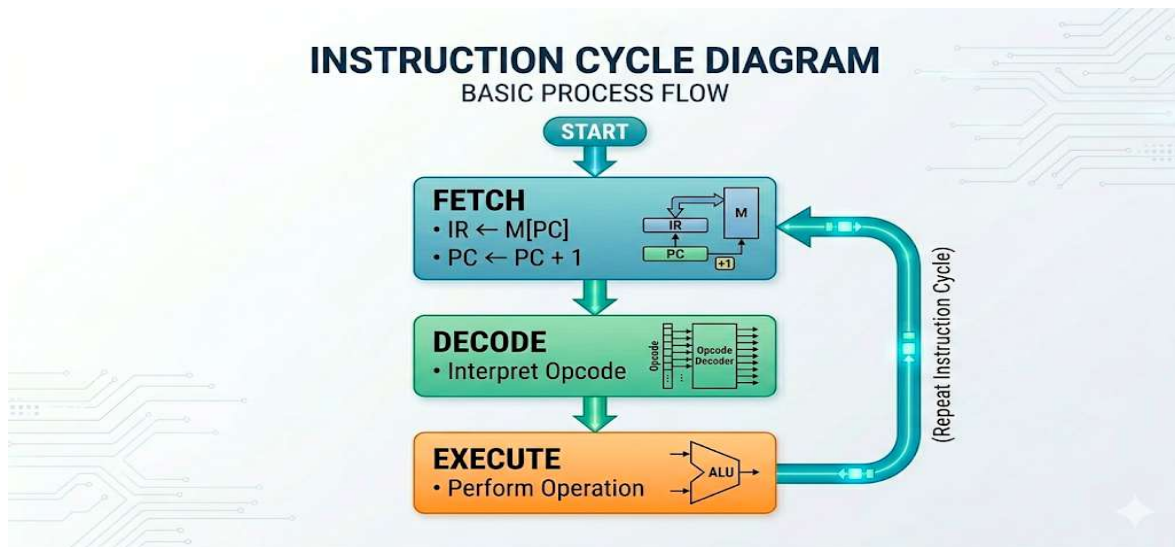
4.4.3 Execute Phase

- The control unit generates the necessary control signals to perform the operation.
- For an arithmetic instruction (e.g., ADD X):
 - Compute effective address of X (may involve AR, index register, etc.).
 - Read operand from memory into DR.
 - Transfer DR to AC (if accumulator is destination) or to ALU.
 - ALU performs addition ($AC + \text{operand}$).
 - Result stored back into AC (or a general-purpose register).
- For a branch instruction (e.g., JMP X):
 - Effective address of X is computed.
 - Transfer X to PC.
- For a store instruction (e.g., STORE X):
 - Compute effective address of X into AR.
 - Transfer AC to DR.
 - Perform memory write.

4.4.4 Store Phase (Write-Back)

- Some instructions write a result back to memory or a register.
- This phase is often considered part of execute.
- For RISC load-store architectures, only LOAD and STORE instructions access memory; ALU results go to registers.

4.4.5 Instruction Cycle Diagram



4.5 CPU Implementation Techniques

The instruction cycle can be implemented in different ways, affecting performance and complexity.

4.5.1 Single-Cycle Implementation

- Each instruction executes in one clock cycle.
- All phases (fetch, decode, execute, write-back) complete within that single cycle.
- The clock cycle time must be long enough for the slowest instruction.
- Advantage: Simple control logic.
- Disadvantage: Inefficient – fast instructions (e.g., register add) wait for the slowest instruction (e.g., memory load). Clock speed is limited.
- Used in very simple processors (e.g., some microcontroller cores).

4.5.2 Multi-Cycle Implementation

- Each instruction is broken into multiple steps (clock cycles), e.g., cycle 1: fetch, cycle 2: decode, cycle 3: execute, cycle 4: write-back.
- Different instructions take different numbers of cycles.
- The clock cycle time can be shorter because each step does less work.
- Advantage: Better resource utilisation; faster overall than single-cycle for most workloads.
- Disadvantage: More complex control unit (must sequence the steps).
- Used in many classic CISC processors and some RISC designs.

4.5.3 Pipelined Implementation

- Multiple instructions are overlapped in execution.
- While one instruction is being executed, the next is being decoded, and the one after that is being fetched.

- Example: In a 4-stage pipeline (Fetch, Decode, Execute, Write-back), up to 4 instructions are in different stages simultaneously.
- Advantage: Increases throughput (instructions per second). One instruction can complete every clock cycle (ideal).
- Disadvantages: Hazards (structural, data, control) require handling; pipeline flush on branches.
- Used in all modern high-performance processors (ARM, x86, RISC-V).

Pipeline diagram (text representation):

Clock cycle	Fetch stage	Decode stage	Execute stage	Write-back stage
1	Inst 1			
2	Inst 2	Inst 1		
3	Inst 3	Inst 2	Inst 1	
4	Inst 4	Inst 3	Inst 2	Inst 1
5	Inst 5	Inst 4	Inst 3	Inst 2

4.6 Data Path and Control Path

4.6.1 Data Path

- The data path is the collection of hardware components that process and move data.
- It includes: registers, ALU, multiplexers (selectors), buses, and memory interface (MAR, MDR).
- The data path is where operands travel, operations are computed, and results are stored.

4.6.2 Control Path

- The control path generates the control signals that tell the data path what to do.
- It includes: the control unit (hardwired or microprogrammed), decoder logic, and sequencing logic.
- Control signals select which register outputs to the bus, which ALU operation to perform, when to read or write memory, etc.

4.6.3 Their Relationship

- The control path does not process data; it only orchestrates.
- The data path does not decide what to do; it only follows control signals.
- Together, they implement the instruction cycle.

Example: For an ADD instruction, the control path sends signals to:

- Enable register R1 output onto bus A.
 - Enable register R2 output onto bus B.
 - Set ALU to ADD.
 - Enable the result bus into register R3.
- Meanwhile, the data path physically moves the bits and performs the addition.

4.7 Summary of Module 4

- The CPU contains the ALU, control unit, and a set of registers.
- Important registers: Accumulator (AC), Address Register (AR), Data Register (DR), Instruction Register (IR), Program Counter (PC), Temporary Register (TR), Input Register (INPR), Output Register (OUTR).
- Registers are connected via a common internal bus to save wiring; only one register can output to the bus at a time.
- The instruction cycle has four phases: fetch, decode, execute, and (implicitly) write-back.
- CPU implementations: single-cycle (simple but slow), multi-cycle (better resource use), and pipelined (high throughput, used in modern CPUs).
- The data path moves and processes data; the control path generates the signals that direct the data path.

Exercises

A. Fill in the Blanks

1. The register that holds the result of an arithmetic operation in a simple computer is called the _____.
2. The _____ register holds the address of the next instruction to be fetched.
3. In a common bus system, only _____ register can output its contents to the bus at any one time.
4. The four phases of the instruction cycle are fetch, decode, execute, and _____.
5. A _____ implementation overlaps multiple instructions to increase throughput.

B. Short Answer Questions

1. What is the purpose of the Temporary Register (TR)? Give an example of when it is used.
2. Explain how the common bus system works. Why is it used instead of direct connections between every pair of registers?
3. Describe the fetch phase step by step, naming the registers involved.

4. Compare single-cycle and pipelined implementations. What is the main advantage of pipelining?
5. What is the difference between the data path and the control path? Who generates the control signals?

C. Diagram Question (Paper or digital)

Draw a simplified diagram of a common bus system connecting five registers: PC, AR, DR, IR, and AC. Label the bus, the output enable lines (tri-state buffers), and the input load lines. Indicate how data flows from PC to AR.

D. True or False

1. The Instruction Register (IR) holds the address of the next instruction. (True / False)
2. In a multi-cycle implementation, every instruction takes the same number of clock cycles. (True / False)
3. The Input Register (INPR) holds data from an input device before it is processed. (True / False)
4. Pipelining can cause hazards such as data dependencies between instructions. (True / False)
5. The control path processes data like the ALU. (True / False)

Quiz (Multiple Choice)

1. Which register is used as the accumulator in a basic computer?
A) PC
B) AC
C) IR
D) AR
2. The register that holds the address of a memory location to be accessed is the:
A) Data Register (DR)
B) Accumulator (AC)
C) Address Register (AR)
D) Instruction Register (IR)
3. In a common bus system, how many registers can drive the bus at the same time?
A) One
B) Two
C) All registers
D) None
4. During the fetch phase, the instruction is moved from memory into which register?
A) PC
B) AR
C) DR then IR
D) AC

5. Which CPU implementation technique completes one instruction per clock cycle (ideally) by overlapping stages?
- A) Single-cycle
 - B) Multi-cycle
 - C) Pipelined
 - D) Microprogrammed
6. The component that generates control signals to direct the data path is the:
- A) ALU
 - B) Control path (control unit)
 - C) Data path
 - D) Accumulator
7. A pipeline hazard caused by a branch instruction is called a:
- A) Data hazard
 - B) Structural hazard
 - C) Control hazard
 - D) Memory hazard
8. In a basic computer, the Output Register (OUTR) holds data that is:
- A) Read from memory
 - B) Sent to an output device
 - C) Received from an input device
 - D) Used as an ALU operand
9. After fetching an instruction, the Program Counter (PC) is:
- A) Reset to zero
 - B) Incremented
 - C) Loaded from the IR
 - D) Saved to memory
10. Which implementation requires the clock cycle to be long enough for the slowest instruction?
- A) Single-cycle
 - B) Multi-cycle
 - C) Pipelined
 - D) Both multi-cycle and pipelined

Quiz Answers

- 1. B – AC (Accumulator).
- 2. C – Address Register (AR) holds memory addresses.
- 3. A – Only one register drives the common bus at a time.
- 4. C – Instruction goes to DR (MDR) then to IR.
- 5. C – Pipelined.
- 6. B – Control path (control unit).
- 7. C – Control hazard (branch).
- 8. B – OUTR holds output data.
- 9. B – PC is incremented to point to the next instruction.
- 10. A – Single-cycle requires one long cycle for all instructions.